9.3.1 Microbial fuel cell(MFC)

A microbial fuel cell is a bio-electrochemical system that drives an electric current by using bacteria and mimicking bacterial interaction found in nature **(Fig. 9.2)**. Microbial fuel cells work by allowing bacteria to oxidize and reduce organic molecules. Bacterial respiration is basically one big redox reaction in which electrons are being moved around. A MFC consists of an anode and a cathode separated by a proton exchange membrane. Microbes at the anode oxidize the organic fuel generating protons which pass through the membrane to the cathode and the electrons pass through the anode to the external circuit to generate current.

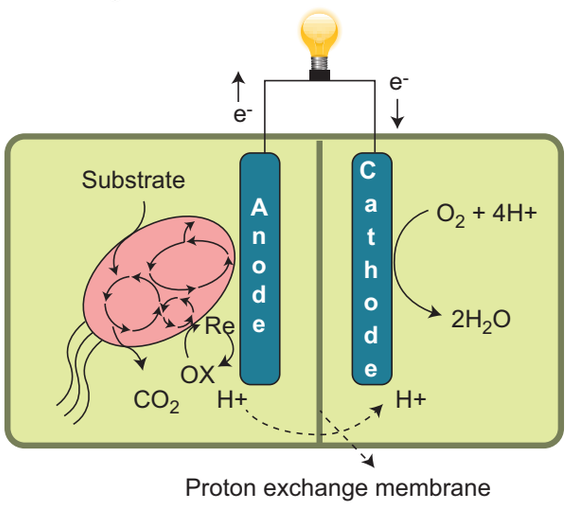


Fig. 9.2 Microbial fuel cell

9.4 Microbes in the production of biogas

Biogas is a mixture of different gases produced by the breakdown of organic matter in the absence of oxygen. Biogas can be produced from raw materials such as agricultural wastes, manure, municipal wastes, plant material, sewage, food waste, etc., Biogas is produced under anaerobic condition, when the organic materials are converted through microbiological reactions into gas and organic fertilizer. Biogas primarily consists of methane (63 percent), along with CO_2 and hydrogen. Methane producing bacteria are called **methanogens** and one such common bacterium is *Methanobacterium*. Biogas is devoid of smell and burns with a blue flame without smoke. The *Methanogens* are also present in anaerobic sludge and rumen of cattle. In rumen, these bacteria help in the breakdown of cellulose. The excreta of cattle called dung is commonly called "**Gobar**". Gobar gas is generated by the anaerobic decomposition of cattle dung. It consists of methane, CO_2 with some hydrogen, nitrogen and other gases in trace amounts.

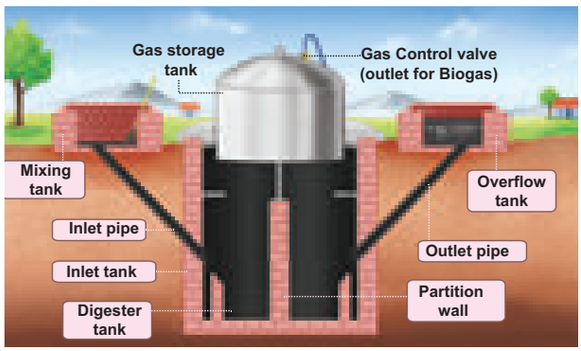


Fig. 9.3 Biogas Unit

In a biogas plant, anaerobic digestion is carried out in an air tight cylindrical tank known as digester **(Fig. 9.3)**. It is made up of concrete bricks and cement or steel. Bio-wastes are collected and slurry of dung is fed into this digester. It has a side opening into which organic materials for digestion are incorporated for microbial activity. Anaerobic digestion is accomplished in three stages: solubilisation, acidogenesis and methanogenisis. The outlet is connected to a pipe to supply biogas. The slurry is drained through another outlet and is used as fertilizer. Biogas is used for cooking and lighting. The technology of biogas production was developed in India mainly due to the efforts of Indian Agricultural Research Institute (IARI) and Khadi and Village Industries Commission (KVIC).

9.5 Bioremediation

The use of naturally occurring or genetically engineered microorganisms to reduce or degrade pollutants is

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